Definition of the Learning Task: Binary Classification of Heart Disease Using a Multilayer Perceptron (MLP)

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1 Problem Definition (Natural Language)

The central objective of this project is to develop and evaluate a machine learning model capable of accurately predicting the **presence or absence of heart disease** in a patient.

Using a set of commonly available clinical and demographic attributes (such as age, sex, resting blood pressure, cholesterol levels, electrocardiogram results, etc.), the system must act as a decision support tool. The model will analyze a patient's profile and output a binary classification: "Disease Present" (1) or "Disease Absent" (0). The purpose is to provide a reliable, early-stage risk prediction that can assist medical professionals in the diagnostic process.

2 Problem Specification

Input Data

A feature vector X representing a single patient.

Description of the Input Data

The input vector X is composed of a set of preprocessed numerical and categorical features. The original dataset (UCI Heart Disease) has been cleaned, imputed, and transformed.

- **Demographic Features:** `age` (numeric), `sex` (binary).
- **Symptomatic Features:** `cp` (chest pain type, one-hot encoded), `exang` (exercise-induced angina, binary).
- **Clinical Measurements:** `trestbps` (blood pressure), `chol` (cholesterol), `fbs` (fasting blood sugar, binary), `restecg` (ECG results, one-hot), `thalach` (max heart rate), `oldpeak` (ST depression), `slope` (ST slope, one-hot), `ca` (major vessels, one-hot), `thal` (thallium state, one-hot).

Preconditions

For the model to accept the input, it must be fully preprocessed as follows:

1. It must not contain any null values (having been imputed with the median for numeric data and the mode for categorical data).
2. All categorical variables (`cp`, `restecg`, `slope`, `ca`, `thal`) must be converted to a numeric format via one-hot encoding.
3. The input vector X must have the exact same number of dimensions as the training data (after one-hot encoding).

4. All features must be **scaled** (standardized to a mean of 0 and a standard deviation of 1) using the `StandardScaler` object that was fitted **only** on the training set.

Output Data

A single binary prediction, y_{pred}.

Postconditions

The output y_{pred} is a value belonging to the set $\{0, 1\}$.

- $y_{pred} = 1$: The model predicts the patient **has** heart disease.
- $y_{pred} = 0$: The model predicts the patient **does not have** heart disease.

Internally, the MLP's output neuron (using a sigmoid activation) produces a probability $p \in [0, 1]$. The final binary output is obtained by applying a threshold (typically 0.5), where $y_{pred} = 1$ if $p \geq 0.5$, and $y_{pred} = 0$ if $p < 0.5$.

3 Specification of the Learning Task (T, P, E)

Following the standard definition of a machine learning problem:

Task (T)

The task is a **Supervised Binary Classification**. Given a patient's feature vector X, the model's task is to learn the mapping function $f(X) \to y$, where y is the class label (0 or 1) indicating the presence of heart disease.

Performance (P)

The primary performance measure will be **Accuracy** on the test set (the percentage of correct predictions on unseen data). Given the nature of the medical problem, where false negatives (failing to detect a disease) can be costly, the following metrics derived from the confusion matrix will also be critically evaluated:

- **Precision**
- **Recall (Sensitivity)**
- **F1-Score**

Experience (E)

The learning experience consists of the processed and scaled training set (X_train_scaled, y_train). This set contains approximately 730 samples (80% of the cleaned 920-patient dataset). Each sample is a pair (feature vector, true class label). The model learns by adjusting its internal weights (parameters) via the **backpropagation** algorithm, aiming to minimize the **binary cross-entropy** loss function on this training data.